

CHAPTER 5

SHALLOW-WATER SUBTIDAL BENTHIC ASSOCIATIONS

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

Much more is known about shallow-water communities than deep-sea communities. This long chapter reflects our substantial knowledge about the sea just below the low-tide mark. Two related environments are not covered in this chapter: Coral reefs, because of their uniqueness, are treated separately (Chapter 9), and although the subtidal and intertidal regions share many features, the intertidal region is covered in Chapter 6. Chapter 5 is subdivided into seven sections and a brief introduction, COVERAGE AND DEFINITIONS, which divides the habitat into its component parts. The first section, ENVIRONMENTAL CONDITIONS, briefly explores the environment of the region. The remainder of the chapter, comprising six sections, examines the various benthic habitats as classified by their dominant features: UNVEGETATED SEDIMENTARY ENVIRONMENTS, ROCKY SUBTIDAL COMMUNITIES, KELP BEDS AND FORESTS, SEA GRASS COMMUNITIES, SOME SPECIAL COMMUNITIES, and BIOLOGY OF POLAR SEAS.

Coverage and Definitions

The region covered in Chapter 5 is delineated by the edge of the continental shelf (ca. < 200 m deep) and the level of the lowest tides.

macrofauna	microfauna	sublittoral
meiofauna	neritic	

- Physically, this zone is divided into two regions: the bottom (soft- and hard-bottom benthos) and its overlying waters (the neritic zone). The organisms swimming above the bottom, crawling over the surface, or burrowing into the substrate are, once again, divided into size categories: macro-, meio-, and microfauna.

I. Environmental Conditions

Wave action, salinity, temperature, light penetration, and food supply all are important factors to benthic organisms in the subtidal region. Each factor influences the benthic community in its own right, but the factors also have synergistic effects.

food supply	light	soft bottom
grain size	long-period ocean swells	storm waves
hard bottom	rocky bottom	temperature
infauna	salinity	wave action

- Subtidal regions with soft bottoms tend to lack topographical relief and are rather meager in their offering of niches to the infauna. Size of grains, chemical composition, and percent of organic material composing the substratum are important in determining species composition. Subtidal regions with hard bottoms, including rocky bottoms, may have considerable vertical relief and therefore numerous habitats.

II. Unvegetated Sedimentary Environments

The communities of the unvegetated subtidal environment are characterized by organisms whose distribution varies in space and time (that is, is patchy). This section is divided into seven parts:

- Sampling
- Infaunal Associations
- Community Pattern and Structure

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- Community Structure and Change
- Positive Interactions
- Vertical Distribution and Competition
- Community Patterns, Stability, and Variability

amensalism	exploitative competition	resettlement
bioturbation	fecal cones	scuba
browsers	grab samples	sediment destabilizers
burrowing predators	grain size	sediment stabilizers
C.G. Joh Petersen	horizontal distribution	side-scan sonar
cage experiments	indirect competition	Smith–McIntyre grab
cannibalism	infauna	soft-sediment communities
commensalism	infauna predators	surface cones
community structure	interference competition	surface predators
competition	mesh size	suspension feeding
competitive interference	parallel bottom communities	temporal mosaic patches
deep-dwelling clams	patch	trophic group amensalism
delayed metamorphosis	Petersen grab	vertical distribution
deposit feeders	pits	wave-induced disturbance
digging predators	recruitment	weasel predators
direct competition	refuge	
epifauna predators	Reineck box corer	

1. Several types of grab samples have been employed to quantitatively sample the shallow marine benthos: the Petersen grab, the Smith–McIntyre grab, and the Reineck box corer. None of these devices provides an intimate look at the benthic environment; that type of examination requires either a good imagination to extrapolate data from grab samples to form a holistic view or a personal visit using either scuba, a submersible, or an ROV with a good video system.
2. The Danish biologist C.G. Petersen, who studied the inshore benthos in European waters, named the benthic communities he found based on the dominant organisms. (N.B.: A few authors have incorrectly cited the spelling of his name as Peterson.)
3. Benthic communities from around the world occurring at the same depth and possessing similar species and sediment types are known as parallel bottom communities.
4. Describing the patterns and structures in benthic communities is not an easy task, given the variation in species distribution, the sheer numbers of macrofauna (> 0.5 mm) present, seasonal variations, etc.
5. Community structure may be described according to the density of each species, or the animals can be lumped into functional groups according to a variety of possible characteristics. For example, one could classify organisms according to their size (for example, microfauna), the way they feed (for example, deposit feeders), where they live (for example, infauna), what they do to the sediments (sediment stabilizer or destabilizer), or their trophic group (for example, herbivores).
6. Abiotic factors influencing community change are clear. They include temperature and wave action. The consequence of the latter is sedimentation or erosion: either burial or exposure, respectively. Biotic factors influencing community are complex. They include competition, predation, and recruitment.
7. Competition among sympatric species can be complex; both direct and indirect competition may be seen.
8. Three types of predators may be seen in the shallow-water benthos, based on their effects on the sediment: surface predators, burrowing predators (including weasel-like predators), and digging predators. The latter two types have a greater effect on the infauna than surface predators because they greatly disturb the sediment. Experiments that use cages to exclude predators provide much insight into the impact of predation on the composition of infaunal organisms.
9. The infauna is not randomly distributed within the sediments. Depending on the location, animals may burrow two meters into the substratum.

10. The inshore benthos is patchy, and the distribution and composition of those patches may change with time. These changes in faunal composition can include some curious interactions. The text-book describes several examples.

III. Rocky Subtidal Communities

While not found everywhere, rocky subtidal communities can be common in certain regions of the world. They are diverse communities composed mainly of low-profile, colonial, encrusting species or gregarious forms, kelp forests, and coral reefs (Chapter 9).

alternate stable states	keystone species	refuge
coloniality	mutualism	role reversal
disease	niche partitioning	space competitors
grazers	overgrowth	storm disturbance
gregarious settlement	patches	sweeper tentacles
kelp forests	recruitment	

1. Co-occurrence of the large number of species in this habitat involves biotic and abiotic factors, including disturbance, space competition, grazing, mass recruitment, mutualistic interactions, and keystone species.
2. This habitat is a world of surfaces (rocks) and biotic surfaces (organisms). There is no depth component as there is in sedimentary communities. However, neither the abiotic surfaces nor the biotic surfaces are fungible as pure surfaces. That is to say, for the larval stages of encrusting organisms, surfaces may be equal in area, but some are “more equal than others.” This condition leads to substratum selection by the larvae.
3. Alternate stable states are small regions or patches where a specific community dominates. In each stable state, a different community is present and remains so until the dominant species is disturbed. At that point, a rearrangement of community structure occurs, with the newly established community being stable until the next disturbance.

IV. Kelp Beds and Kelp Forests

The kelp community can significantly alter what would otherwise be a relatively monotonous landscape. Kelps provide an important element that really does make habitats akin to a terrestrial forest, with several vertical strata as well as horizontal complexity. This section is divided into two parts:

- Structure and Distribution
- Ecology and Life Cycle

alternation of generations	grazing	pneumatocyst (float)
blade	holdfast	production
canopy	kelp	recruitment window
community structure	kelp beds	sporophyte
El Niño	kelp forests	stipe
ENSO	keystone predators	storms
float	La Niña	understory algae
food web	life cycle	zonation
gametophyte	mechanical damage	

1. A canopy and an understory are found in kelp forests. Kelps must receive adequate light, have a hard surface to which to attach their holdfasts, and have the proper temperature and nutrient levels before significant growth can take place.
2. Growth in kelps originates from tissues at the level of the holdfast, so loss of stipe and blade does not kill the plant. Their life cycle is an alternation of generations between an asexual sporophyte, which is the macroscopic kelp plant, and a sexual gametophyte, which is microscopic.
3. Net annual productivity of kelp beds (ca. 800–2,000 g C/m²) can be several times the production of phytoplankton on a same-area basis.

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4. The population dynamics of a kelp forest are complex, involving the interaction of both biotic (grazing, recruitment of the macrophytic sporophytic stage, and recruitment and loss of keystone predators) and abiotic (wave action and mechanical damage, light levels, nutrient concentrations, and water temperature) forces.

V. Seagrass Communities

Although comprising only about 50 species, seagrasses are of unparalleled ecological importance. Their productivity is truly remarkable. This section is divided into six parts:

- Composition and Distribution
- Environmental Conditions
- Productivity
- Structure and Biological Interactions
- Relationships to Other Systems
- Ecology

anaerobic environments	nutrient cycling	stubble
climax community	pioneer stage	succession
currents	refuge	temperature
desiccation	rhizome	terminal stage
detritus	roots	<i>Thalassia</i>
dominance	seagrass	turbidity
eelgrass	sediment trap	turtle grass
erosion	sere	wasting disease
grazing halos	shelter	wave action
<i>Halodule</i>	stabilization	<i>Zostera</i>

1. Seagrasses are exceedingly productive, forming dense carpets across the bottom sediments. These plants are often colonized by various epiphytic plants and animals.
2. These vascular plants are affected by water and sediment movement, light and turbidity, heat and cold, desiccation, and anaerobic conditions. They also are affected by grazing, disease, and competition for space and light.
3. The productivity of seagrasses is remarkable, both in temperate waters (500–1,000 g C/m²/yr) and in tropical waters (> 4,000 g C/m²/yr). The dynamics of seagrass nutrient cycling are similar to what is found in midwestern prairies.
4. Seagrasses are important to inshore waters. While their primary production is high, they are a food source for relatively few grazers. Actually, in a sense, they are more like forested terrestrial ecosystems than like terrestrial grasses. In both cases (seagrasses and terrestrial forests), the detrital food component is strong.
5. Seagrasses are important to sediment stabilization; they are anchored by their dense root system, and the blades trap additional sediments, thus creating a depositional environment. The blades also provide cover for small animals, thereby acting as a refuge.

VI. Some Special Communities

This brief section discusses two unusual communities: the sea-pen (*Ptilosarcus*) community and the sea-pansy (*Renilla*) community.

Cormier, M.J., and W.W. Lorenz. 1996. Isolated renilla luciferase and method of use thereof. *Biotechnol. Adv.* 14: 294–295.

Weightman, J.O., and D.J. Arsenault. 2002. Predator classification by the sea pen *Ptilosarcus gurneyi* (Cnidaria): role of waterborne chemical cues and physical contact with predatory sea stars. *Can. J. Zool.* 80: 185–191.

VII. Biology of Polar Seas

Although both are dominated by cold water and extensive ice fields, the Arctic and Antarctic regions are fundamentally different. This section is divided into three parts:

- Physical Conditions of the Arctic and Antarctic Regions
- Biological Comparisons of Arctic and Antarctic Seas
- Ecological Communities of Polar Seas

anchor ice	enclosed sea	salp
annual sea ice	endemic	sea ice
Antarctic Convergence	Fram Strait	seed population
anticyclonic currents	hard-bottom community	shelf
Arctic Ocean basin	ice scour	soft-bottom community
Beauford gyre	ice shelf water	subice platelet layer
Bering Strait	impoverished fauna	Thorson's rule
disturbance-associated fauna	krill	upwelling
diurnal vertical migration (DVM)	McMurdo Sound	Zones I, II, and III
	pack ice	

1. The Arctic Ocean is a cold sea dominated by a perennial layer of ice called pack ice. The land masses surrounding the Arctic Ocean have extensive shallow shelves. Thus, this deep basin is isolated from the Pacific and Atlantic Oceans, with only shallow connections of the Bering Strait (70 m) and the Fram Strait (400 m), respectively. In contrast, the Antarctic is an ice-covered land mass surrounded by a circumpolar ring of water. Here, the continental shelf is narrow, rapidly dropping into deep water. Table 5.1 and Figure 5.42 illustrate the differences between these two regions.
2. Wave action is of much less importance to benthic organisms in the polar regions than elsewhere. Here, anchor ice forms on the bottom and, if it lifts off the bottom, may carry organisms to the sea ice, where they become entrapped. Thus, in the polar regions, ice motion replaces wave energy as a significant abiotic factor in structuring the benthic community. The depth to which anchor ice extends is about 30 m.
3. Thorson's Rule—that cold-water benthic organisms should exhibit a predominance of nonpelagic development—does not appear to hold in these waters. In the Antarctic, pelagic planktotrophic development is common.
4. The Antarctic possesses a greater benthic species diversity and greater endemism than the Arctic. The Arctic fauna has a lower species diversity, and much of the Arctic's life-forms are of Atlantic origin.
5. Sea ice provides the unique habitat of an ice community, which does not appear to have a freshwater parallel. This community has a well-developed microbial food web. (To some workers, this group of organisms does not compose a true community.)
6. South of the Antarctic Convergence, the dominant euphausiid shrimp is *Euphausia superba*, but in the Arctic Ocean, no such dominating species occurs.
7. Hard-bottom communities are more common than soft-bottom communities in the Antarctic, but both have been studied. Most soft-bottom communities appear to be structured by competition, but a rare type of soft-bottom community has three zones (canopy, understory, and subsurface). It appears that this community is regulated by small predatory crustaceans in the understory. In the hard-bottom community of the McMurdo Sound, there are three zones. At 0 to 15 m of depth, ice scouring permits only mobile organisms to survive. Between 15 to 30 m of depth, anchor ice begins to lessen its controlling influence; here, cnidarians, especially anemones, dominate. Finally, at depths greater than 30 m, anchor ice is no longer present; here, sponges dominate, but the regions possess a diverse fauna as well.

ENRICHMENT EXERCISES

Shallow-Water Subtidal Benthic Ecology

Two selected Web sites dealing with the shallow-water subtidal benthos, active as of April 3, 2004, are as follows:

- Biosedimentology Unit Homepage: <http://www.gla.ac.uk/Acad/IBLS/Biosed/main.htm>
- National Marine Fisheries Service Web site: <http://www.nefsc.noaa.gov/>

(N.B.: A word of caution: If you search for the keyword “kelp” on any Internet search engine, you will be overwhelmed with hits for companies that sell products that in some way relate to or use kelp.)

Anholt, B.R. 1990. An experimental separation of interference and exploitative competition in a larval damselfly. *Ecology* 71: 1483–1493. [While its subject is not marine and certainly not benthic, this paper addresses the concepts of interference and exploitative competition.]

Aronson, R.B., and D.B. Blake. 2001. Global climate change and the origin of modern benthic communities in Antarctica. *Am. Zool.* 41: 27.

Cerrano, C., et al. 1999. Organism-quartz interactions in structuring benthic communities: towards a marine bio-mineralogy? *Eco. Lett.* 2: 1–4.

Dayton, P.K., M.J. Tegner, and P.B. Edwards. 1999. Temporal and spatial scales of kelp demography: the role of oceanographic climate. *Ecol. Monogr.* 69: 219–250.

Dernie, K.M., M.J. Kaiser, and R.M. Warwick. 2003. Recovery rates of benthic communities following physical disturbance. *J. An. Eco.* 72: 1043–1057.

Dial, R., and J. Roughgarden. 1998. Theory of marine communities: the intermediate disturbance hypothesis. *Ecology* 79: 1412–1424.

Dybas, C.L. 1993. Sequoia forest beneath the sea. *Wildlife Conser.* 96: 24–35.

Estes, J.A., and D.O. Duggins. 1995. Sea otters and kelp forests in Alaska: generality and variation in a community ecological paradigm. *Ecol. Monogr.* 65: 75–100.

Gaylord, B., and M.W. Denny. 1997. Flow and flexibility I: effects of size, shape and stiffness in determining wave forces on the stipitate kelps *Eisenia arborea* and *Pterygophora californica*. *J. Exp. Bio.* 200: 3141–3164.

Harrold, C., K. Light, and S. Lisin. 1998. Organic enrichment of submarine-canyon and continental-shelf benthic communities by macroalgal drift imported from nearshore kelp forests. *Limno. Oceanogr.* 43: 669–678.

MacIsaac, H.J., and J.J. Gilbert. 1991. Discrimination between exploitative and interference competition between Cladocera and *Keratella cochlearis*. *Ecology* 72: 924–937. [While its subject is not marine and certainly not benthic, this paper addresses the two concepts of interference and exploitative competition.]

Steinberg, P.D., J.A. Estes, and F.C. Winter. 1995. Evolutionary consequences of food chain length in kelp forest communities. *PNAS* 92: 8145–8148.

Utter, B.D., and M.W. Denny. 1996. Wave-induced forces on the giant kelp *Macrocystis pyrifera* (Agardh): field test of a computational model. *J. Exp. Bio.* 199: 2645–2654.